INDUSTRIAL INTERNET OF THINGS

External Viva Question Bank with Answers

Fourth Year • Artificial Intelligence and Data Science
Savitribai Phule Pune University • 2020 Course

Complete syllabus coverage — Units I to VI

Conceptual questions • Technical questions • Comparisons • Working principles • Applications • Industry 4.0 and 5.0

Prepared for external practical/oral examination revision

How to use this question bank

Read each answer until you can explain it naturally in your own words. For comparison questions, remember the point of difference and one example. For protocol questions, focus on the purpose, range/power/data-rate characteristics where specified by your syllabus, and the application. An external examiner may ask follow-up questions based on any keyword in your answer.

Coverage overview

- UNIT I — INTRODUCTION TO INDUSTRIAL INTERNET OF THINGS — 20 questions
- UNIT II — IIoT SYSTEM PROTOCOLS — 26 questions
- UNIT III — IIoT ARCHITECTURE — 22 questions
- UNIT IV — CLOUD AND DATA ANALYTICS FOR IIoT — 23 questions
- UNIT V — IIoT SECURITY CHALLENGES AND SOLUTIONS — 24 questions
- UNIT VI — APPLICATIONS, USE CASES AND INDUSTRY REVOLUTION — 28 questions

Total: 143 viva questions with answers.

Important note: Some syllabus-listed technologies have vendor-specific or implementation-dependent specifications. The answers describe their commonly taught characteristics; verify exact numerical specifications against your faculty notes if your examiner expects a particular version or regional deployment.

UNIT I — INTRODUCTION TO INDUSTRIAL INTERNET OF THINGS

20 questions • Answer each in clear, complete sentences

Q1. What is the Internet of Things (IoT)?

Answer: IoT is a network of physical objects embedded with sensors, software and communication capabilities. These objects collect and exchange data over the internet or another network, allowing monitoring, automation and intelligent decisions.

Q2. What is Industrial Internet of Things (IIoT)?

Answer: IIoT is the use of connected sensors, machines, control systems and software in industrial environments. It collects real-time operational data to improve productivity, quality, safety, reliability and maintenance.

Q3. What is the main objective of IIoT?

Answer: The main objective is to connect industrial assets and processes so that data can be monitored and analysed in real time. This helps industries reduce downtime, improve efficiency, predict failures and make data-driven decisions.

Q4. Explain the history and evolution of IIoT.

Answer: IIoT evolved from industrial automation, embedded systems, machine-to-machine communication and traditional SCADA systems. The growth of low-cost sensors, wireless networks, cloud computing, big-data analytics and AI enabled machines to become connected and intelligent. The term Industrial Internet gained prominence through industry initiatives in the early 2010s.

Q5. Differentiate between IoT and IIoT.

Answer: IoT is used across consumer and general-purpose applications such as smart homes and wearables. IIoT focuses on industrial operations such as manufacturing, energy and logistics. IIoT generally requires higher reliability, security, accuracy, availability and real-time response because failures can affect production, equipment or human safety.

Q6. What are the four industrial revolutions?

Answer: Industry 1.0 introduced mechanisation using water and steam power. Industry 2.0 introduced electricity and mass production through assembly lines. Industry 3.0 introduced electronics, computers and automation. Industry 4.0 connects machines and systems using IoT, cyber-physical systems, cloud and analytics. Industry 5.0 adds a stronger focus on human-centricity, sustainability and resilience.

Q7. What is the role of IIoT in an industry?

Answer: IIoT connects machines, people and industrial software. It supports real-time monitoring, process automation, quality control, asset tracking, energy management, predictive maintenance and better decision-making.

Q8. How is IIoT used in manufacturing processes?

Answer: Sensors monitor machine speed, temperature, vibration, pressure and product quality. Data is sent to gateways and industrial platforms for analysis. The system can alert operators, adjust process parameters or schedule maintenance, improving throughput and reducing defects.

Q9. What are the main requirements of an IIoT system?

Answer: Important requirements include reliable connectivity, accurate sensing, interoperability, scalability, low latency where needed, data storage and processing, cybersecurity, availability, maintainability and integration with existing industrial systems.

Q10. What design considerations are important when implementing IIoT?

Answer: Consider the industrial environment, sensor range and accuracy, communication protocol, network coverage, latency, power consumption, safety, cybersecurity, data volume, scalability, interoperability, cost and compatibility with legacy equipment.

Q11. What is plant maintenance?

Answer: Plant maintenance is the set of activities used to keep industrial equipment and facilities in safe, reliable working condition. It includes inspection, servicing, repair and replacement of components.

Q12. How does IIoT improve plant maintenance?

Answer: IIoT sensors continuously monitor equipment conditions such as vibration, temperature, pressure and current. Analytics can identify abnormal patterns and predict possible failures, allowing maintenance teams to act before breakdowns occur.

Q13. Differentiate preventive and predictive maintenance.

Answer: Preventive maintenance is performed at planned time or usage intervals, whether or not a fault is developing. Predictive maintenance uses condition data and analytics to estimate when equipment may fail, so maintenance can be performed when needed.

Q14. What is condition-based maintenance?

Answer: Condition-based maintenance is maintenance triggered by the actual measured condition of equipment rather than a fixed calendar schedule. For example, a motor may be serviced when vibration exceeds a defined threshold.

Q15. What are the key benefits of IIoT?

Answer: Benefits include reduced downtime, higher productivity, improved product quality, lower maintenance cost, better asset utilisation, energy savings, improved worker safety, supply-chain visibility and faster decisions.

Q16. What are the major challenges in IIoT implementation?

Answer: Challenges include cybersecurity risks, integration with legacy machines, lack of common standards, high initial investment, connectivity limitations, handling large data volumes, skills shortages, data privacy and ensuring reliable operation.

Q17. What are the key opportunities offered by IIoT?

Answer: Opportunities include smart factories, remote asset monitoring, predictive maintenance, automated quality inspection, energy optimisation, digital twins, connected supply chains and new data-driven services.

Q18. Enlist applications of IIoT.

Answer: Applications include smart manufacturing, predictive maintenance, oil and gas monitoring, power-grid management, smart energy meters, water treatment, mining, logistics, process industries, connected vehicles, agriculture, healthcare equipment and building management.

Q19. What is a smart factory?

Answer: A smart factory is a manufacturing facility in which machines, sensors, software and people are digitally connected. Production data is monitored and used to coordinate processes, improve quality and respond quickly to changes.

Q20. Give a simple real-life example of IIoT.

Answer: A motor in a factory has vibration and temperature sensors. The readings are sent to an analytics platform. If vibration rises abnormally, the system alerts maintenance staff before the motor fails, preventing an unexpected production stoppage.

UNIT II — IIoT SYSTEM PROTOCOLS

26 questions • Answer each in clear, complete sentences

Q1. What is an industrial sensor?

Answer: An industrial sensor detects a physical quantity such as temperature, pressure, flow, level, vibration, proximity or humidity and converts it into a measurable signal for a control or monitoring system.

Q2. What is an actuator?

Answer: An actuator converts a control signal into physical action. Examples include motors, valves, relays, pneumatic cylinders and hydraulic actuators.

Q3. Differentiate sensors and actuators.

Answer: A sensor measures a physical condition and provides input to a system. An actuator receives a control command and produces an output action. For example, a temperature sensor measures heat while a valve actuator controls fluid flow.

Q4. What are the roles of sensors in IIoT?

Answer: Sensors collect real-time process and machine data. They enable condition monitoring, process control, fault detection, quality inspection, energy monitoring and predictive maintenance.

Q5. What are the roles of actuators in IIoT?

Answer: Actuators execute decisions made by controllers or automation systems. They can open or close valves, start or stop motors, move robotic arms and adjust industrial process parameters.

Q6. What is an IIoT sensor network?

Answer: An IIoT sensor network is a group of connected sensors that measure industrial conditions and communicate their readings to gateways, controllers or platforms. It may use wired or wireless communication.

Q7. What is process automation?

Answer: Process automation uses control systems, sensors, actuators and software to perform industrial operations with limited manual intervention. It improves consistency, safety, speed and productivity.

Q8. What is data acquisition in IIoT?

Answer: Data acquisition is the process of collecting signals from sensors and machines, converting them into usable digital data, and sending that data to a monitoring or analytics platform.

Q9. What is a gateway in an IIoT sensor network?

Answer: A gateway connects field devices or local networks to higher-level networks or cloud platforms. It may translate protocols, aggregate data, filter information and provide security functions.

Q10. What is ZigBee?

Answer: ZigBee is a low-power, low-data-rate wireless communication technology based on IEEE 802.15.4. It supports mesh networking and is suitable for sensor and control networks.

Q11. State the characteristic features of ZigBee.

Answer: ZigBee is designed for low power consumption, low data rates, low cost and short-range communication. It supports star, tree and mesh topologies and can connect many nodes. It is commonly used for monitoring and control rather than high-bandwidth data.

Q12. What is ZigBee IP?

Answer: ZigBee IP is an IPv6-based version of the ZigBee networking approach. It enables IP connectivity for low-power wireless devices and helps connect sensor networks with IP-based infrastructure.

Q13. What is Z-Wave?

Answer: Z-Wave is a low-power wireless communication technology mainly used in control and automation networks. It operates in sub-GHz frequency bands, which can reduce interference with crowded 2.4 GHz systems. It is commonly associated with smart-home devices, but illustrates low-power mesh networking.

Q14. What is Wi-Fi Backscatter?

Answer: Wi-Fi Backscatter is a communication approach in which a device transmits data by reflecting or modulating existing radio-frequency signals rather than generating a full new radio signal. It can greatly reduce the energy needed by very low-power devices.

Q15. What is NFC?

Answer: Near Field Communication is a short-range wireless technology operating at 13.56 MHz. It enables communication over a few centimetres and is commonly used for contactless identification, pairing and access systems.

Q16. What is 6LoWPAN?

Answer: 6LoWPAN stands for IPv6 over Low-Power Wireless Personal Area Networks. It allows small, resource-constrained devices on IEEE 802.15.4 networks to communicate using IPv6 through header compression and adaptation mechanisms.

Q17. What is RPL?

Answer: RPL stands for Routing Protocol for Low-Power and Lossy Networks. It is an IPv6 routing protocol designed for networks where devices have limited power, memory and processing capability and links may be unreliable.

Q18. What is Sigfox?

Answer: Sigfox is a low-power wide-area network (LPWAN) technology designed for small amounts of data over long distances. It uses very narrow-band radio communication, low data rates and low energy consumption, making it suitable for battery-powered sensors.

Q19. What is nWave?

Answer: In this syllabus context, nWave refers to a low-power wide-area wireless approach intended for long-range, low-data-rate sensor communication. Its focus is low energy use and infrequent small messages. Exact operating bands and service specifications depend on the particular implementation.

Q20. What is DASH7?

Answer: DASH7 is a wireless sensor-network protocol associated with the ISO/IEC 18000-7 standard. It operates in sub-GHz bands, supports low-power communication and is suited to telemetry, asset tracking and sensor applications.

Q21. What is Low-Power Wi-Fi?

Answer: Low-Power Wi-Fi refers to Wi-Fi approaches or modes designed to reduce energy use in connected devices. It aims to retain IP-based connectivity while reducing radio-on time and power consumption for suitable IoT applications.

Q22. What is LTE Category-M (LTE-M)?

Answer: LTE-M, also called LTE Cat-M1, is a 3GPP cellular LPWAN technology for IoT. It supports wide-area coverage, mobility and relatively higher data rates than very narrow-band LPWAN systems, while using power-saving features such as PSM and eDRX.

Q23. What is RPMA?

Answer: RPMA stands for Random Phase Multiple Access. It is a proprietary LPWAN technology associated with Ingenu, designed for long-range, low-power IoT connectivity using spread-spectrum techniques in the 2.4 GHz band.

Q24. Compare ZigBee and Wi-Fi.

Answer: ZigBee is designed for low-power, low-data-rate sensor and control networks and supports mesh topologies. Wi-Fi provides much higher data rates and IP connectivity but generally consumes more power. ZigBee suits battery-operated sensors; Wi-Fi suits cameras, gateways and devices needing higher bandwidth.

Q25. Why are LPWAN technologies used in IIoT?

Answer: LPWAN technologies provide long-range connectivity with low energy consumption and low-cost communication for devices that send small amounts of data. They are useful for remote meters, tanks, pipelines and widely distributed sensors.

Q26. What factors decide the choice of a wireless protocol?

Answer: Important factors are range, data rate, power consumption, latency, reliability, interference, network size, mobility, security, cost, spectrum availability and compatibility with existing infrastructure.

UNIT III — IIoT ARCHITECTURE

22 questions • Answer each in clear, complete sentences

Q1. What are the basic components of an IIoT system?

Answer: Common components include sensors, actuators, controllers, gateways, routers, modems, communication networks, cloud brokers, servers, databases, analytics software and user applications.

Q2. What is the function of a sensor in IIoT architecture?

Answer: A sensor measures a physical parameter and generates data. It forms the field-level input of the architecture and provides information about the machine or process.

Q3. What is the function of a gateway?

Answer: A gateway collects data from field devices, performs protocol conversion, filtering or aggregation, and forwards information to enterprise networks or cloud services. It can also send control commands back to devices.

Q4. What is a router?

Answer: A router forwards data packets between different IP networks. It selects a path based on routing information and helps connect industrial networks to other networks.

Q5. What is a modem?

Answer: A modem modulates and demodulates signals to enable data transmission over a communication medium. In IIoT deployments, it may provide cellular, DSL or other network connectivity.

Q6. What is a cloud broker?

Answer: A cloud broker is an intermediary software component or service that helps devices and applications exchange messages with cloud services. It may handle message routing, queuing, protocol mediation and device connectivity.

Q7. What is an IIoT server?

Answer: An IIoT server receives and processes device data, manages applications or databases, and provides services such as device management, analytics, dashboards and APIs.

Q8. What is a Wireless Sensor Network (WSN)?

Answer: A WSN is a network of spatially distributed sensor nodes that monitor physical or environmental conditions and communicate their readings, often wirelessly, to a sink or gateway.

Q9. What are the main elements of a sensor node?

Answer: A typical sensor node contains a sensing unit, processor or microcontroller, communication transceiver and power source. It may also include memory and an actuator interface.

Q10. Explain the basic flow of data in an IIoT system.

Answer: Sensors collect data from machines. A controller or gateway gathers and may preprocess it. The network sends it to an edge or cloud platform, where servers store and analyse it. Dashboards show results and alerts or control commands can be sent back to actuators.

Q11. What is IIoT architecture?

Answer: IIoT architecture is the structured arrangement of devices, networks, computing platforms, data services and applications that work together to connect industrial assets and deliver monitoring, analytics and control.

Q12. What is a business model in IIoT?

Answer: An IIoT business model explains how an organisation creates, delivers and captures value using connected industrial technology. Examples include selling connected equipment, remote monitoring, predictive-maintenance services or usage-based services.

Q13. What is a reference architecture?

Answer: A reference architecture is a general blueprint that identifies common components, layers, functions and relationships in a system. It guides the design of specific IIoT solutions and improves consistency and interoperability.

Q14. What is the Industrial Internet Architecture Framework (IIAF)?

Answer: The Industrial Internet Architecture Framework is an architecture framework developed by the Industrial Internet Consortium. It provides viewpoints and guidance for describing industrial internet systems, including business, usage, functional and implementation concerns.

Q15. What is the purpose of IIAF?

Answer: IIAF helps stakeholders understand complex industrial systems, identify requirements and concerns, organise system functions and communicate architecture decisions. It is a framework for describing architectures, not a single fixed product design.

Q16. Explain the IIoT sensing layer.

Answer: The sensing layer interacts with the physical industrial environment. It includes sensors and data acquisition devices that measure parameters such as temperature, vibration, pressure, flow and speed.

Q17. Explain the IIoT processing layer.

Answer: The processing layer handles collected data through filtering, aggregation, storage, edge computing, cloud computing and analytics. It converts raw measurements into useful information.

Q18. Explain the IIoT communication layer.

Answer: The communication layer transfers data and commands between devices, gateways, edge systems and cloud platforms using wired or wireless technologies and suitable protocols.

Q19. Explain the IIoT networking layer.

Answer: The networking layer provides connectivity and data routing among industrial devices, local networks, enterprise systems and remote platforms. It supports addressing, routing, segmentation and secure communication.

Q20. Differentiate edge computing and cloud computing.

Answer: Edge computing processes data close to the machines or sensors, reducing latency and bandwidth use. Cloud computing processes and stores data in remote data centres and offers scalable resources. IIoT systems often combine both approaches.

Q21. Why is interoperability important in IIoT?

Answer: Interoperability allows devices and software from different vendors to exchange and understand data. It reduces integration difficulty, avoids isolated systems and supports expansion of industrial deployments.

Q22. What is scalability in IIoT architecture?

Answer: Scalability is the ability of an IIoT system to handle growth in devices, data volume, users or applications without unacceptable loss of performance or reliability.

UNIT IV — CLOUD AND DATA ANALYTICS FOR IIoT

23 questions • Answer each in clear, complete sentences

Q1. What is cloud computing?

Answer: Cloud computing provides computing resources such as servers, storage, databases, networking and software over a network on demand. It allows organisations to scale resources without owning all the underlying infrastructure.

Q2. What is Cloud of Things (CoT)?

Answer: Cloud of Things refers to the integration of IoT devices and cloud computing. Devices collect data while cloud platforms provide storage, device management, processing, analytics and application services.

Q3. What is Predix?

Answer: Predix is an industrial IoT software platform developed by GE for connecting industrial assets, collecting operational data and building industrial applications and analytics solutions.

Q4. What is PTC ThingWorx?

Answer: ThingWorx is an industrial IoT application platform from PTC. It supports connecting devices and systems, modelling industrial assets, building applications, visualising data and developing connected-product solutions.

Q5. What is Microsoft Azure in IIoT?

Answer: Microsoft Azure is a cloud platform that provides services for device connectivity, data ingestion, storage, analytics, AI, security and application development. Azure IoT services can be used to build and manage connected industrial solutions.

Q6. What are cloud services in IIoT?

Answer: Cloud services provide remote infrastructure, platforms and software for device connectivity, data storage, processing, analytics, visualisation, application hosting, identity management and security.

Q7. What is SaaS?

Answer: Software as a Service provides ready-to-use software over the internet, usually through a browser or application. The provider manages the application and underlying infrastructure. Example: a hosted industrial monitoring dashboard.

Q8. What is PaaS?

Answer: Platform as a Service provides a managed environment for developers to build, test, deploy and manage applications without managing the underlying servers and operating systems. Example: a cloud application-development platform.

Q9. What is IaaS?

Answer: Infrastructure as a Service provides virtualised computing resources such as virtual machines, storage and networking. The customer manages operating systems and applications, while the provider manages the physical infrastructure.

Q10. Differentiate SaaS, PaaS and IaaS.

Answer: SaaS delivers a complete application to the user. PaaS delivers a managed platform for building and deploying applications. IaaS delivers virtual computing infrastructure. The customer's management responsibility is generally highest in IaaS and lowest in SaaS.

Q11. What is data analytics in IIoT?

Answer: Data analytics in IIoT is the process of examining industrial data to discover patterns, detect abnormal conditions, understand performance and support decisions about machines and processes.

Q12. Why is data analytics important in IIoT?

Answer: Industrial devices generate large volumes of data. Analytics turns that data into actionable insights for predictive maintenance, quality improvement, process optimisation, energy efficiency, fault detection and planning.

Q13. What are the types of data analytics?

Answer: Descriptive analytics explains what happened. Diagnostic analytics investigates why it happened. Predictive analytics estimates what may happen. Prescriptive analytics recommends what action should be taken.

Q14. What is data visualisation?

Answer: Data visualisation is the graphical presentation of data using charts, graphs, gauges, maps and dashboards. It helps operators identify trends, compare values and understand system status quickly.

Q15. Name common data visualisation techniques used in IIoT.

Answer: Common techniques include line charts for trends, bar charts for comparisons, scatter plots for relationships, histograms for distributions, heat maps for intensity, gauges for live values, dashboards for KPIs and geographic maps for asset locations.

Q16. What is a KPI?

Answer: A Key Performance Indicator is a measurable value used to assess how effectively a process or organisation is achieving an objective. Industrial KPIs include production rate, downtime, defect rate, energy per unit and Overall Equipment Effectiveness.

Q17. What is a Digital Twin?

Answer: A Digital Twin is a digital representation of a physical asset, process or system that is updated using data from its real-world counterpart. It can be used to monitor, simulate, analyse and optimise the physical system.

Q18. Why is a Digital Twin needed?

Answer: A Digital Twin helps understand asset behaviour, test scenarios without disturbing production, predict failures, optimise performance, improve product design and support lifecycle management.

Q19. What are the elements of a Digital Twin?

Answer: Typical elements include the physical object or process, its digital model, sensors and data connections, data storage, analytics or simulation tools, and a feedback mechanism linking insights to the physical system.

Q20. Explain the Digital Twin process.

Answer: First identify the physical asset and its purpose. Build a digital model using design and operational information. Connect sensors and data sources. Synchronise real-time or periodic data. Analyse and simulate behaviour, then use the findings to improve the physical asset or process.

Q21. What information is required to design a Digital Twin?

Answer: Requirements include asset specifications, geometry or process model, operating conditions, sensor measurements, historical data, maintenance records, boundary conditions, relationships between components and the intended use cases.

Q22. Differentiate a simulation and a Digital Twin.

Answer: A simulation models how a system may behave under selected assumptions and inputs. A Digital Twin is connected to a real physical asset and is updated with its operational data. A Digital Twin may use simulations, but connectivity and ongoing data synchronisation distinguish it.

Q23. What is a hybrid edge-cloud IIoT platform?

Answer: A hybrid edge-cloud platform divides work between local edge devices and cloud services. Time-critical processing and local control can happen near machines, while the cloud provides large-scale storage, fleet-wide analytics and application services.

UNIT V — IIoT SECURITY CHALLENGES AND SOLUTIONS

24 questions • Answer each in clear, complete sentences

Q1. Why is security important in IIoT?

Answer: IIoT systems control or monitor critical industrial operations. A cyberattack can interrupt production, damage equipment, expose confidential information or create safety risks. Security protects people, processes, assets and data.

Q2. What is cybersecurity?

Answer: Cybersecurity is the practice of protecting systems, networks, devices and data from unauthorised access, misuse, disruption, modification or destruction.

Q3. How is conventional web technology related to IIoT?

Answer: IIoT uses internet and web technologies such as IP networking, web APIs, messaging protocols, cloud services and dashboards. However, industrial systems also have real-time, availability and safety requirements that ordinary web applications may not face.

Q4. What are the major vulnerabilities of IIoT?

Answer: Vulnerabilities include weak passwords, insecure device firmware, unpatched software, exposed network services, poor encryption, insecure APIs, default credentials, physical access, legacy protocols and inadequate network segmentation.

Q5. What is privacy in IIoT?

Answer: Privacy concerns the proper collection, use, storage and sharing of information. Industrial data may reveal production volumes, operational methods, employee activity or sensitive business information.

Q6. What are the security requirements of IIoT?

Answer: Key requirements include confidentiality, integrity, availability, authentication, authorisation, accountability, privacy, secure updates, resilience and protection of physical processes.

Q7. What is the CIA triad?

Answer: The CIA triad represents Confidentiality, Integrity and Availability. Confidentiality prevents unauthorised disclosure, integrity prevents unauthorised alteration, and availability ensures systems and data remain accessible when needed.

Q8. What is threat analysis?

Answer: Threat analysis identifies assets, possible attackers, attack paths, vulnerabilities, likelihood and impact. It helps prioritise security controls according to the risk to the industrial operation.

Q9. What is identity establishment in IIoT security?

Answer: Identity establishment assigns and verifies a unique identity for a device, user or service. Certificates, cryptographic keys and secure credentials can be used so that only trusted entities communicate.

Q10. What is access control?

Answer: Access control determines which authenticated users or devices can access particular resources and what actions they are permitted to perform. It follows the principle of giving only the permissions needed.

Q11. What is message integrity?

Answer: Message integrity ensures that data has not been altered during transmission or storage. Cryptographic hash functions and message authentication codes can help detect tampering.

Q12. What is non-repudiation?

Answer: Non-repudiation provides evidence that a particular entity performed an action or sent a message, so it cannot credibly deny that action later. Digital signatures and secure audit logs can support it.

Q13. What is availability in IIoT security?

Answer: Availability means that devices, networks and services remain operational and accessible when required. Redundancy, backups, fault tolerance, monitoring and denial-of-service protection help maintain availability.

Q14. What is authentication?

Answer: Authentication verifies the identity of a user, device or service. Common methods include passwords, certificates, cryptographic keys, tokens and multi-factor authentication.

Q15. Differentiate authentication and authorisation.

Answer: Authentication answers 'Who are you?' by verifying identity. Authorisation answers 'What are you allowed to do?' by checking permissions after identity has been established.

Q16. What is encryption?

Answer: Encryption converts readable data into an encoded form using a cryptographic key. Only an authorised party with the correct key can decrypt it. It protects confidentiality during storage or transmission.

Q17. What is network segmentation?

Answer: Network segmentation divides a network into smaller zones to limit unnecessary communication. In industry, separating IT networks from operational technology (OT) networks can reduce the spread and impact of attacks.

Q18. Name common network security techniques for IIoT.

Answer: Techniques include firewalls, network segmentation, VPNs, secure protocols, encryption, intrusion detection and prevention, access control, certificate-based authentication, secure configuration and continuous monitoring.

Q19. What is a firewall?

Answer: A firewall monitors and controls network traffic according to security rules. It can block unauthorised connections and restrict communication between industrial network zones.

Q20. What is an intrusion detection system (IDS)?

Answer: An IDS monitors network or host activity for suspicious behaviour or known attack patterns and generates alerts. An intrusion prevention system can also take action to block detected threats.

Q21. What is the difference between IT and OT security?

Answer: IT security focuses mainly on protecting information and business systems. OT security protects systems that monitor or control physical industrial processes. OT places especially strong emphasis on safety, availability and predictable operation.

Q22. What are the management aspects of cybersecurity?

Answer: Management aspects include security policies, asset inventories, risk assessments, staff training, incident response plans, vendor management, patch management, audits, backup planning and continuous improvement.

Q23. Why are legacy industrial systems difficult to secure?

Answer: Legacy systems may use old protocols, unsupported software and equipment that cannot be easily patched or restarted. They may also lack modern authentication or encryption, so upgrades must be planned carefully to avoid production and safety risks.

Q24. What is the principle of least privilege?

Answer: It means giving each user, device or service only the minimum permissions required to perform its task. This limits damage if an account or device is compromised.

UNIT VI — APPLICATIONS, USE CASES AND INDUSTRY REVOLUTION

28 questions • Answer each in clear, complete sentences

Q1. What is smart robotics?

Answer: Smart robotics combines robots with sensors, connectivity, control software and sometimes AI. Robots can monitor their condition, adapt to tasks, coordinate with other systems and share operational data.

Q2. What is smart metering?

Answer: Smart metering uses connected meters to measure and communicate consumption data automatically. It enables remote readings, usage monitoring, fault detection and more informed energy or water management.

Q3. What is smart irrigation?

Answer: Smart irrigation uses soil-moisture, temperature, humidity or weather data to control water delivery. It can operate pumps and valves automatically, reducing water waste and improving crop management.

Q4. What is a smart factory?

Answer: A smart factory is a digitally connected production environment in which machines, products, people and systems exchange data. It supports flexible production, automation, quality monitoring and optimisation.

Q5. How is IIoT used in healthcare service industries?

Answer: IIoT connects medical devices and hospital equipment for remote patient monitoring, equipment tracking, cold-chain monitoring, predictive maintenance of devices and improved resource management. Patient-data privacy and safety are essential.

Q6. What is a smart office?

Answer: A smart office uses connected lighting, HVAC, occupancy sensors, access systems and energy meters to improve comfort, security, space utilisation and energy efficiency.

Q7. What is smart logistics?

Answer: Smart logistics uses connected vehicles, GPS, RFID, sensors and analytics to track goods and vehicles, monitor storage conditions, optimise routes and improve supply-chain visibility.

Q8. What is a Cyber-Physical System (CPS)?

Answer: A Cyber-Physical System integrates physical processes with computation and networking. Sensors observe the physical world, software analyses information, and control actions influence the physical process.

Q9. Differentiate IoT and Cyber-Physical Systems.

Answer: IoT emphasises connecting physical objects and exchanging data. CPS emphasises tight integration between computation, communication and physical processes, often including feedback and control. Many industrial systems are both IoT systems and CPS.

Q10. What is Industry 4.0?

Answer: Industry 4.0 is the fourth industrial revolution. It refers to the integration of digital technologies such as IIoT, cyber-physical systems, cloud computing, AI, robotics and data analytics into industry to create connected and intelligent production.

Q11. Why is Industry 4.0 important now?

Answer: Affordable sensors, faster networks, cloud platforms, AI, robotics and large-scale data processing have matured. Industries need greater efficiency, flexibility, quality, traceability and resilience, making connected digital operations practical and valuable.

Q12. What are the main characteristics of Industry 4.0?

Answer: Key characteristics include connectivity, interoperability, information transparency, decentralised decision-making, real-time data, automation, flexibility, modularity and integration across the value chain.

Q13. What are the design principles of Industry 4.0?

Answer: Commonly stated design principles are interoperability, virtualisation, decentralisation, real-time capability, service orientation and modularity. They guide the design of connected, flexible and intelligent systems.

Q14. Explain interoperability as an Industry 4.0 principle.

Answer: Interoperability means machines, devices, people and systems can connect and exchange information. It allows different components to work together across the production environment.

Q15. Explain virtualisation in Industry 4.0.

Answer: Virtualisation creates a digital representation of physical processes or machines using sensor data and models. It helps operators observe and understand the real system through a digital view.

Q16. Explain decentralisation in Industry 4.0.

Answer: Decentralisation allows connected systems or machines to make certain decisions locally instead of relying on one central controller for every action. This can improve responsiveness and flexibility.

Q17. Explain real-time capability in Industry 4.0.

Answer: Real-time capability means collecting, processing and responding to data quickly enough to meet the needs of the industrial process. The required response time depends on the application.

Q18. Explain service orientation in Industry 4.0.

Answer: Service orientation means making system functions and data available as services through standard interfaces, so other applications or systems can use them.

Q19. Explain modularity in Industry 4.0.

Answer: Modularity means production systems can be expanded, rearranged or reconfigured by adding or changing modules. It supports flexible manufacturing and changing product requirements.

Q20. What are the advantages of Industry 4.0?

Answer: Advantages include higher productivity, improved quality, reduced downtime, flexible production, better traceability, lower operating costs, improved safety and faster decisions based on data.

Q21. What are applications of Industry 4.0?

Answer: Applications include smart factories, autonomous robots, predictive maintenance, digital twins, connected supply chains, additive manufacturing, smart energy systems, automated inspection and mass customisation.

Q22. What is Industry 5.0?

Answer: Industry 5.0 is an approach that complements Industry 4.0 by placing human well-being, sustainability and resilience at the centre of industrial development. It promotes collaboration between people and advanced technologies.

Q23. What is Society 5.0?

Answer: Society 5.0 is a human-centred vision of a super-smart society in which digital technologies such as AI, IoT, robotics and big data are integrated into physical and social spaces to address societal challenges.

Q24. Differentiate Industry 4.0 and Industry 5.0.

Answer: Industry 4.0 mainly emphasises connectivity, automation, data and smart production. Industry 5.0 builds on these technologies but places greater emphasis on human-centricity, sustainability, resilience and collaboration between humans and machines.

Q25. What is human-centricity in Industry 5.0?

Answer: Human-centricity means designing technology and work systems around human needs, skills, safety and well-being. Technology should support and empower workers rather than simply replace them.

Q26. What is sustainability in Industry 5.0?

Answer: Sustainability means reducing environmental impact and using resources responsibly. Examples include energy-efficient production, waste reduction, circular-economy practices and responsible material use.

Q27. What is resilience in Industry 5.0?

Answer: Resilience is the ability of an industry to anticipate, withstand, adapt to and recover from disruptions such as supply shortages, cyber incidents or equipment failures.

Q28. Give an example combining IIoT and Industry 4.0.

Answer: In a smart factory, machine sensors send vibration and temperature data to an edge gateway. Analytics predicts equipment failure, a dashboard alerts the maintenance team and production planning adjusts the schedule. This demonstrates connectivity, analytics and intelligent operations.